



## Enhancement of biomass and bioethanol production using *Chlorococcum minutum* with the assistance of nano-MnO<sub>2</sub>

P. Gnaneswari<sup>a</sup>, B. Sourya Swetha<sup>b</sup>, K. Riazunnisa<sup>c</sup> , V. Anu Prasanna<sup>d</sup>, M. Rajasekhara<sup>e</sup>,  
E.C. Surendranatha Reddy<sup>b</sup>, M. Narendra Reddy<sup>f</sup> , L. Subramanyam Sarma<sup>f</sup> ,  
T. Chandrasekhar<sup>a,\*</sup>

<sup>a</sup> Department of Environmental Science, Yogi Vemana University, Kadapa, 516005, A.P., India

<sup>b</sup> Department of Biotechnology, Yogi Vemana University, Kadapa, 516005, A.P., India

<sup>c</sup> Department of Biotechnology & Bioinformatics, Yogi Vemana University, Kadapa, 516005, A.P., India

<sup>d</sup> Department of Zoology, Yogi Vemana University, Kadapa, 516005, A.P., India

<sup>e</sup> Department of Zoology, Sri Venkateswara University, Tirupati, 517502, A.P., India

<sup>f</sup> Department of Chemistry, Yogi Vemana University, Kadapa, 516005, A.P., India

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### ABSTRACT

Algae offer a transformative solution for sustainable biofuel production, addressing key energy and environmental challenges. The current investigation evaluated the effects of manganese dioxide (MnO<sub>2</sub>) nanomaterials on the growth, chlorophyll (biomass) and bioethanol contents using the green microalgae, *Chlorococcum minutum*. Tris-acetate-phosphate (TAP) medium with different concentrations of nano-MnO<sub>2</sub> was used for *C. minutum* cultures. TAP with 0.1 mg/L of nano-MnO<sub>2</sub> achieved 29.40 mg/L of total chlorophyll, compared to other treated and control cultures. Similarly, a high amount of dry algal biomass (160 mg/L) was noticed with 0.1 mg/L of nano-MnO<sub>2</sub> treatment. Maximum ethanol yield i.e. 30.37 g/L, was noticed with the feedstock grown in TAP with 0.1 mg/L of nano-MnO<sub>2</sub> for 48 h. However, high nano-MnO<sub>2</sub> concentrations reduced both growth and ethanol production, highlighting the importance of optimal dosage. To the best of our knowledge, this is the first report on the role of nano-MnO<sub>2</sub> in bioethanol production from algae and the obtained results provide ample scope for sustainable output. Overall, nano-MnO<sub>2</sub> may be considered as one of the potential candidates with respect to ethanol production using algal species for commercialization as well as for future research.

### 1. Introduction

Ever-increasing energy requirement with population explosion and incessant non-renewable fossil fuel depletion, along with the negative environmental consequences of fossil fuel usage, demands us to focus on renewable clean energy sources [1,2]. In addition, energy demand across the transport and industrial sectors is high when compared to other sectors [3]. Common renewable energy sources in use are solar energy, wind energy, hydro energy, bioenergy etc. Biomass-based energy is an upcoming form of renewable source which is mainly produced from organic matter derived mostly from plants, microbial organisms including algae, animals and their wastes [3,4]. Biofuels are further classified into bioethanol, biodiesel, biohydrogen etc., depending on the end product available in liquid and gaseous states. Though bioenergy research is emerging, it still needs a lot of effort for full-fledged

commercial practice [5]. Ethanol or bioethanol is a secondary fuel that is helpful for blending purposes with petrol or diesel. Moreover, blending of petrol or diesel with ethanol reduces the pollution levels [6]. In contrast, production of ethanol through chemical methods and also using other organisms and their waste under anaerobic conditions is insufficient. Therefore, focus on the enhancement of bioethanol generation at an industrial scale from photosynthetic life forms such as algal species has gathered momentum due to its environmental friendly process [7].

Algae, including microalgae, have emerged as a prospective alternative for tackling the energy concerns due to their flexible habitat, rigorous growth and high productivity [8]. Algae are divided into green, blue-green, brown and red groups based on their pigment content and the first two groups are being extensively used for the generation of value-added products, including ethanol [9]. One such species,

\* Corresponding author. Department of Environmental Science, Yogi Vemana University, Kadapa, 516005, Andhra Pradesh, India.

E-mail address: [tcsbiotech@gmail.com](mailto:tcsbiotech@gmail.com) (T. Chandrasekhar).

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*C. minutum*, which belongs to green microalgae, stands out from the other species due to its high growth and ability to make biofuels [10]. The rate of bioethanol production from algal species completely depends on the quantity of biomass. Generally, algal biomass consists of rich in carbohydrates with a lower quantity of lignin, which is useful for the production of bioethanol [11]. In contrast, algae and their squander are considered as biowaste in certain places. Therefore, utilization of algal species for the generation of various commercial products, including biofuels, is one of the waste management practices [10,12]. Biofuel content from algae can be further improved by adding certain nutritional components, metals, vitamins and nanoparticles to the algal medium [12]. Further, using the algal feedstock for bioethanol production through fermentation is the most widely used traditional method of alcohol production, which consumes less energy [6,13,14].

Manganese is required as a trace element for the growth of both advanced and primitive plants, including algae. Availability of manganese in soils depends on its different oxidation states. In general,  $Mn^{2+}$  is soluble and oxides of manganese are insoluble in soils [15].  $MnO_2$  is involved in carbohydrate and nitrate metabolic pathways in biological organisms. Specifically, it is involved in the water-splitting stage and oxygen evolution at photosystem II in photosynthesis and is also involved in  $CO_2$  fixation [16]. In contrast,  $MnO_2$  causes toxicity in high doses and inhibits nutrient uptake and conversion will be affected by inhibiting the plasma membrane ATPase translocation activity [17]. Long exposure of green alga, *Chlorella sorokiniana* to  $MnO_2$  leads to growth inhibition, specifically decrease in chlorophyll content was noticed [17]. In addition, high concentrations of  $MnO_2$  cause oxidative stress in algal cells. However, it is also involved in the ROS scavenging process through SOD. Further, environmental conditions such as alteration in temperature and pH will also be affected by the presence of  $MnO_2$ . Nanoparticles are currently used in almost all sectors and are promising applications in the energy sector [18]. Generally, nanomaterials are available as nanowires, nanotubes, nanorods, etc., and are involved in many biological functions [19]. Particularly, manganese dioxide ( $MnO_2$ ) nanoparticles have unique features that make them useful in many kinds of applications, including energy storage, catalysis, water treatment, environmental remediation and are also involved in

many electrochemical and biochemical reactions [20]. By considering both the advantages and disadvantages of  $MnO_2$ , the current investigation aimed to determine the effects of nano- $MnO_2$  on growth and biomass accumulation in *C. minutum* under *in vitro* conditions, ultimately ethanol production levels during fermentation. Bioethanol production was carried out through improved hydrolysis by immobilization of cellulase onto  $MnO_2$  nanoparticles using agricultural waste was reported earlier by Cherian et al. [21]. To the best of our knowledge, this is the first report on bioethanol production from algal species with the assistance of nano- $MnO_2$ .

## 2. Materials and methods

### 2.1. Algal culture collection

In the present work, *C. minutum* was collected from the University of Madras, Chennai, India and the species was already confirmed. The algal strains were preserved using glycerol stocks through the standard protocols to ensure viability and long-term usability. The laboratory has already established an *in vitro* system for algal cultures, which includes media preparation, culture initiation and maintenance, algal feedstock collection, biomass estimation, enzymatic hydrolysis, fermentation and ethanol estimation. Fig. 1 explains the growth and ethanol production using nano- $MnO_2$ .

### 2.2. Nanoparticles synthesis and characterization

Rod-like  $MnO_2$  nanoparticles were synthesized using a hydrothermal method in acidic conditions by adapting the procedure of Yu et al. [22] with minor modifications. Initially, 0.55g of  $KMnO_4$  was dissolved in 25 mL of water under vigorous stirring and 0.89g of  $MnSO_4 \cdot H_2O$  was dissolved in 35 mL of water separately. Later,  $MnSO_4 \cdot H_2O$  solution was added to the  $KMnO_4$  solution dropwise under continuous stirring. Afterwards, 0.5 mL of concentrated HCl was added dropwise under stirring and later this solution was transferred to 200 mL of stainless-steel Autoclave Teflon liner (Indfurr-Superheat Furnaces) and heated at 145 °C for 10 h. Later, it was allowed to reach room temperature and

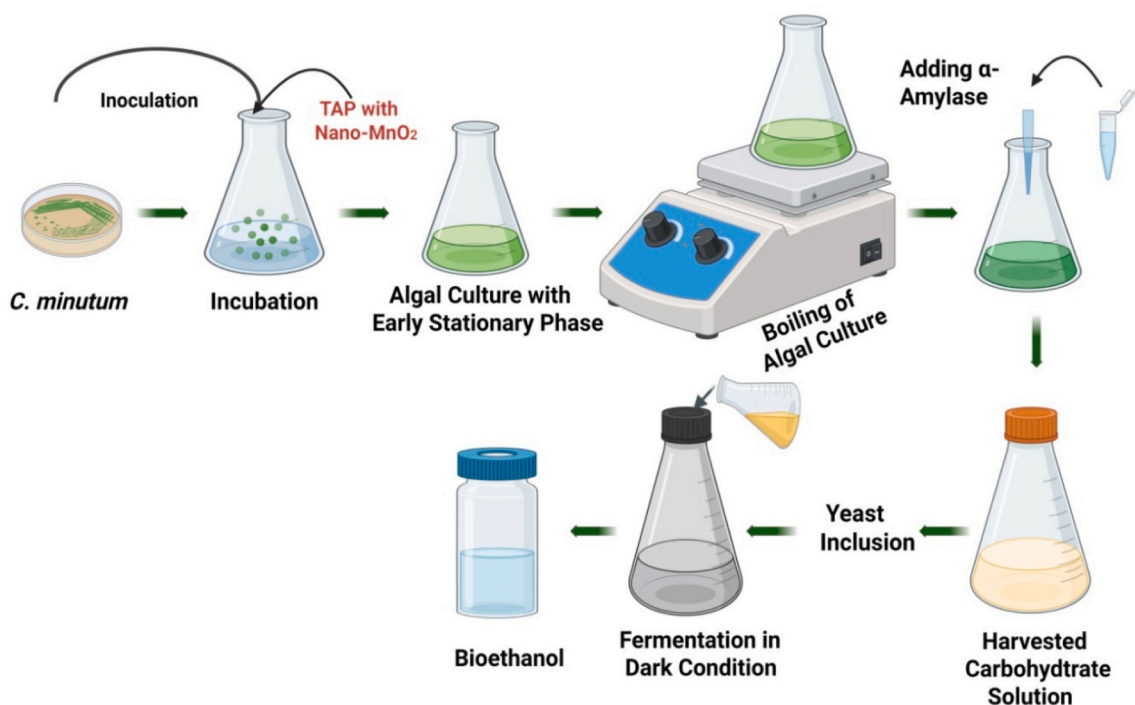


Fig. 1. Schematic representation of ethanol production from algae using nano- $MnO_2$ .

washed with distilled water and absolute ethanol. Finally, the product was collected by centrifugation and dried at 65 °C for 18 h.

To assess the phase structure and crystallite size of MnO<sub>2</sub> nanorods, X-ray diffraction (XRD) measurements were performed. XRD patterns were recorded with a (Rigaku MiniFlex 600) X-ray diffractometer with Cu K<sub>α</sub> radiation ( $\lambda = 1.5406 \text{ \AA}$ ). The morphological details of MnO<sub>2</sub> nanorods were investigated by a (JEOL-JSM-IT500) scanning electron microscope (SEM) operated at an accelerated voltage of (150 kV). The presence of elements and compositions of MnO<sub>2</sub> nanorods was determined using energy dispersive X-ray spectroscopy (EDX) using (APEX-AMETEX) EDX detector attached to SEM. The length and diameter of MnO<sub>2</sub> nanorods were determined by image-J software.

### 2.3. Media preparation and culture initiation

In the current investigation, tris-acetate-phosphate (TAP) medium was used, which includes TAP salts (NH<sub>4</sub>Cl, MgSO<sub>4</sub>·7H<sub>2</sub>O and CaCl<sub>2</sub>·2H<sub>2</sub>O), phosphate solution (KH<sub>2</sub>PO<sub>4</sub> and K<sub>2</sub>HPO<sub>4</sub>), Hutner's solution (Na<sub>2</sub>EDTA·2H<sub>2</sub>O, ZnSO<sub>4</sub>·7H<sub>2</sub>O, H<sub>3</sub>BO<sub>3</sub>, MnCl<sub>2</sub>·4H<sub>2</sub>O, FeSO<sub>4</sub>·7H<sub>2</sub>O, CoCl<sub>2</sub>·6H<sub>2</sub>O, CuSO<sub>4</sub>·5H<sub>2</sub>O and (NH<sub>4</sub>)<sub>6</sub> Mo<sub>7</sub>O<sub>24</sub>·4H<sub>2</sub>O), tris (hydroxymethyl) aminomethane buffer and acetic acid. Different tris-acetate-phosphate (TAP) media were prepared, supplemented with varying concentrations of manganese dioxide (MnO<sub>2</sub>) nanoparticles (0.0, 0.1, 0.3, 0.9 and 2.7 mg/L). The rationale behind the selection of concentrations of nano-MnO<sub>2</sub> is based on their properties and literature. After including the nano-MnO<sub>2</sub>, the flasks were mixed well for dispersion of nanoparticles. pH of all media was adjusted to around 7.0 using a calibrated digital pH meter (Elico Limited, India). Media preparation was carried out in sterilized conical flasks, ensuring at least 2/3 of the flask volume remained empty to provide a gaseous phase for algal growth. Prepared media were sterilized via autoclaving (Inlab Equipment, India) at 121 °C and 15 lbs/in<sup>2</sup> for 20 min. Following sterilization, all the media were cooled to ambient temperature and transferred to a laminar air flow (LAF) chamber (Hitech Products, Chennai, India). Under sterilized conditions, all the media were inoculated with *C. minutum* and all the cultures were incubated in an orbital shaker (ROTEK-LES, India) at 25 °C under continuous light conditions to optimize algal growth. Continuous light was used to maximize the total amount of light energy absorbed for photosynthesis to enhance growth and biomass productivity.

### 2.4. Estimation of chlorophyll contents and dry algal biomass

Chlorophyll content was measured for all the algal cultures during the logarithmic growth phase using a modified process of Arnon's [23] and Varaprasad et al. [10]. Initially, cultures were harvested and transferred into Eppendorf tubes. All the samples were centrifuged at 10,000 rpm for 10 min at 25 °C, and the supernatant was removed. The remaining wet biomass was used to determine chlorophyll contents. Later, 1 ml of 80 % acetone is added to the pellet, vortexed and then placed in the dark for 10 min and later centrifuged at 5000 rpm for 10 min. Further, the supernatant of all the samples were collected and finally, O.D. (optical density) values were estimated using a UV-visible spectrophotometer (Shimadzu UV-1800) and total chlorophyll content was calculated using the standard formula.

Dry algal biomass or dry cell weight (DCW) was estimated using the algal feedstocks grown in different media assisted with various MnO<sub>2</sub> concentrations as per the modified method of Zhu and Lee [24]. Algal cells were harvested by centrifugation at 5000 rpm for 10 min at 25 °C and then washed twice with distilled water to remove tap media components and the resultant biomass was then placed in a hot air oven at 50 °C overnight. The DCW was gravimetrically determined using a bench-top digital weighing balance (Essae Teraoka Pvt. Ltd., India) by following the standard formula (dry algal biomass = post oven dried biomass plus vial-pre oven dried vial weight)

### 2.5. Enzymatic hydrolysis

Algal cultures were harvested at the early stationary phase and subjected to enzymatic hydrolysis. Cells were boiled at 100 °C for 90 min on a hot plate, and then cooled to room temperature. Subsequently, 0.6 g/L of  $\alpha$ -amylase (Sigma Aldrich) was added to the samples and incubated for 90 min by following the methods described by Sulfari et al. [25] and Varaprasad et al. [10] and activity was expressed in U/ml. After hydrolysis, the mixture was centrifuged at 5000 rpm for 10 min at 5 °C. Further, the supernatant was collected, filter-sterilized and stored in sterile tubes. This sterilized supernatant was used as a feedstock for subsequent fermentation processes. As routine work, the concentration of reducing sugars was estimated using the dinitrosalicylic acid (DNS) method.

### 2.6. Fermentation

Yeast peptone medium was prepared by combining 1 % yeast extract, 2 % peptone and 20 % glucose. To this medium, 5.0 g of commercially available baker's yeast (*Saccharomyces cerevisiae*) was dissolved in 50.0 ml of warm water. The solution was prepared under aseptic conditions within a laminar air flow chamber and sealed with aluminium foil. The culture was incubated in an orbital shaker at 120 rpm and 27 °C for 24 h to ensure adequate yeast propagation. Fermentation was conducted in 250-ml Erlenmeyer flasks containing 50 ml of algal feedstock solution supplemented with 10 % yeast culture and maintained at pH 5.5 [10]. To create anaerobic conditions, O<sub>2</sub> was purged using nitrogen gas in the flask before sealing. Cultures were incubated in an orbital shaker at 120 rpm and 27 °C under dark fermentation for optimal mixing and ethanol analysis was carried out at different time intervals.

### 2.7. Ethanol estimation and statistical work

Ethanol levels were estimated using a modified potassium dichromate method, as described by Varaprasad et al. [10]. In the fermentation process, 1.0 ml of potassium dichromate and 5.0 ml of concentrated H<sub>2</sub>SO<sub>4</sub> were added to the samples and readings were taken at 0, 12, 24, 48, 72 and 96 h. Ethanol concentration was determined using a UV-spectrophotometer at a wavelength of 660 nm. A standard curve was generated using commercially available ethanol to quantify the ethanol levels in the experimental samples. ANOVA (one-way analysis of variance) was used to examine whether different nano-MnO<sub>2</sub> concentrations significantly influenced the ethanol production levels at different time intervals. Further, post-hoc comparisons was done using Tukey's test. Both ANOVA and Tukey's tests were carried out by IBM SPSS software version 20 Bar graph and a Heat map were produced by using GraphPad Prism version 8.0.2 [26].

## 3. Results and discussion

### 3.1. Synthesis and characterization of nanoparticles

Production of bioethanol using algal species is one of the best alternatives instead of using edible crop plants [27,28]. Exploring the role of nano-metal oxides on several biochemical pathways of algal species is one of the advanced methods for biofuel production in recent times. Manganese dioxide nanoparticles are used for several purposes [29]. In the present work, the synthesis of MnO<sub>2</sub> nanorods and their effects on the growth and development of *C. minutum* in turn, ethanol production through fermentation were measured.

XRD profile showed the prominent peaks at  $2\theta$  values of 14.17°, 28.73°, 37.44°, 41.02°, 56.70° and 59.43° corresponding to (110), (310), (121), (420), (002) and (260) respectively of  $\alpha$ -MnO<sub>2</sub> phase [JCPDS file No: 044-0141]. The crystallite size of synthesized MnO<sub>2</sub> nanorods was estimated using the Debye-Scherrer equation i.e.  $D = (K\lambda)/(\beta \cos\theta)$ ,  $D$  = Crystallite size (nm),  $K$  = Shape factor (0.9),  $\lambda$  = X-ray

wavelength ( $1.5406 \text{ \AA}$  for Cu  $K_{\alpha}$  radiation),  $\beta$  = Full width at half maximum (FWHM) in radians and  $\theta$  = Bragg angle. The corresponding crystallite size associated with the most intense peak of the (310) plane yielded is found to be  $23.4 \text{ nm}$ . In addition, the average crystallite size across all the selected peaks was found to be  $29.6 \pm 4.0 \text{ nm}$ , indicating uniform nanoscale dimensions suitable for high surface area applications (Fig. 2). Similarly, the laboratory generated the reduced graphene oxide-supported platinum-ruthenium (Pt-Ru/RGO) nanoparticles for bioethanol production from algal species [10].

The  $\text{MnO}_2$  nanorods were examined using a scanning electron microscope to analyze their morphology and microstructure. Fig. 2 shows low-magnification images revealing a dense, even dispersion on the substrate. Nanorods appear uniformly aligned with minimal aggregation and consistent shape. Most observed structures are rod-like, with no significant impurities or alternative forms. Fig. 2 also offers a high-magnification view of individual nanorods with smooth surfaces. Each rod maintains a uniform diameter and length, confirming structural integrity. Dimensions range from approximately  $110\text{--}120 \text{ nm}$  in length and  $30\text{--}35 \text{ nm}$  in diameter. Surface defects and amorphous deposits are negligible, indicating the merit of synthesis. Overall, the SEM observations from Fig. 2 clearly demonstrate the successful synthesis of high-quality  $\text{MnO}_2$  nanorods with homogeneous morphology and well-

defined dimensions. These characteristics are essential for advanced applications in catalysis, electrochemical energy storage, and other related fields. The presence of Mn and O in nano- $\text{MnO}_2$  can be noticed from the EDX profile shown in the bottom side of Fig. 2.

### 3.2. Estimation of chlorophylls and dry algal biomass

Algal cultures grown in TAP with nano- $\text{MnO}_2$  medium enhanced the growth and chlorophyll contents (Fig. 3). The highest total chlorophyll content ( $29.40 \text{ mg/L}$ ) was observed in *C. minutum* grown in TAP medium with  $0.1 \text{ mg/L}$  of nano- $\text{MnO}_2$ . Probably, alterations in biochemical pathways are the reasons for a higher amount of biomass in nano- $\text{MnO}_2$ -assisted algal cultures [16,21]. Specifically, a modification in photosynthetic pathways i.e. photolysis of water at photosystem II, may be the reason for a high amount of chlorophyll content as mentioned by Alejandro et al. [16].

Conversely, other media exhibited the lower chlorophyll contents, indicating growth suppression at higher nanoparticle concentrations. Particularly, TAP with  $2.7 \text{ mg/L}$  of nano- $\text{MnO}_2$  exhibited the lowest total chlorophyll contents and the culture appeared in light blackish green colour due to more amount of nano- $\text{MnO}_2$  in the TAP medium (Fig. 3). Similarly, a decrease in chlorophyll content was observed in

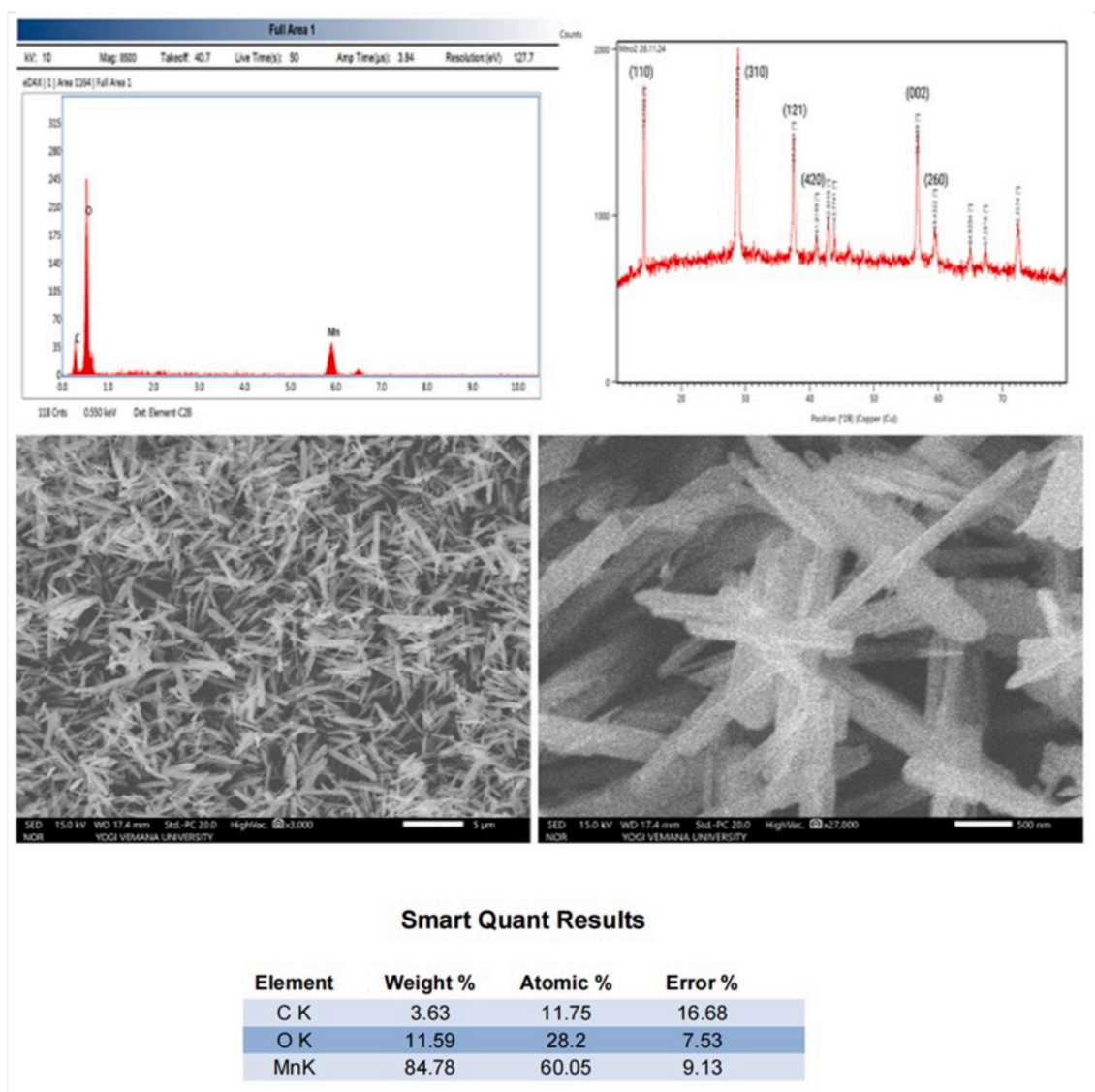


Fig. 2. Characterization of  $\text{MnO}_2$  nanorods using XRD, SEM and EDAX.



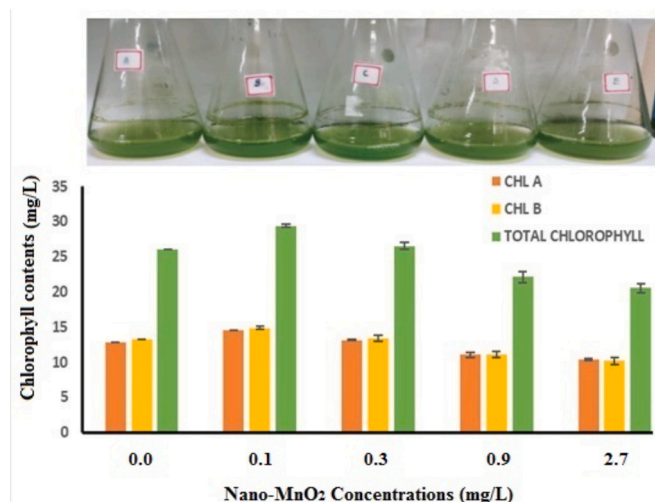


Fig. 3. Growth and chlorophyll content of *C. minutum* in nano-MnO<sub>2</sub> treatments.

*Chlorella sorokiniana* due to nano-MnO<sub>2</sub> toxicity, which leads to growth inhibition [17]. Moreover, algal cultures grown in all the media reached the stationary phase within three days, but cultures grown in TAP with 2.7 mg/L medium exhibited slower growth, reaching the stationary phase after five days. This data indicates that higher concentrations of MnO<sub>2</sub> nanorods reduce the growth in this species, indicating potential inhibitory effects. The detailed chlorophyll content data are summarized in Fig. 3.

Algal feedstock grown in TAP with 0.1 mg/L nano-MnO<sub>2</sub> exhibited high dry biomass (160 mg/L), indicating that this concentration is optimal for algal growth (Fig. 4). In contrast, algal feedstock dry biomass declined progressively and reached a low (60 mg/L) at a 2.7 mg/L dose, when compared with feedstock grown without nano-MnO<sub>2</sub> (120 mg/L). On the whole, 0.1 mg/L of nano-MnO<sub>2</sub> appears to be the most favourable concentration for biomass accumulation and high concentrations exhibit toxicity (or) metabolic stress [16,17].

### 3.3. Bioethanol production from algal feedstocks

Bioethanol production was achieved using the supernatants of enzymatic hydrolysis (reducing sugars) through yeast-mediated dark fermentation. Few fermentation works were carried out in the laboratory for bioethanol production using algal feedstocks [10,13,27,30]. The potassium dichromate technique was used to measure ethanol levels throughout a time period ranging from 0 to 96 h. In the heat map, control and different concentrations of nano-MnO<sub>2</sub> are on the x-axis and

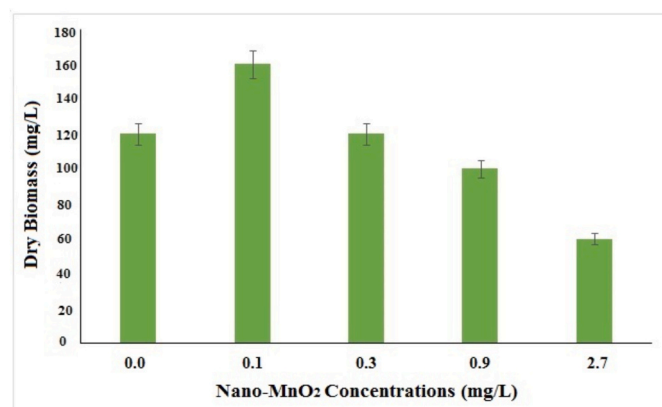


Fig. 4. Dry biomass (DCW) of *C. minutum* in nano-MnO<sub>2</sub> treatments.

time intervals (0, 12, 24, 48, 72 and 96 h) are shown on the y-axis. Each cell in the heat map contains an ethanol value, with different colour intensity. Blue colour indicates the lower ethanol values and red colour indicates higher ethanol values (Fig. 5). The colour progression across the heat map reflects the changes in ethanol production levels for each condition and time interval.

ANOVA results showed a significant difference in bioethanol production at varied time intervals [ $F(5,24) = 4.323$ ,  $P < 0.05$ ]. Additionally, Tukey's test revealed that the yield at 0.1 mg/L concentration was significantly different from other concentrations. Further, the yield at 48 h was significantly different from other time intervals (Fig. 5).

Feedstock of *C. minutum* grown in 0.1 mg/L of nano-MnO<sub>2</sub> resulted highest amount of ethanol (30.37 g/L) production after 48 h of fermentation. Most likely, nano-MnO<sub>2</sub> has a role in the augmentation of carbohydrate production in the photosynthetic pathway by enhancing the electro and biochemical reactions as a catalyst, which in turn results in more ethanol yield [16,31]. Other time intervals in the heat map showed the low levels of ethanol content. Particularly, feedstock grown in a higher concentration of MnO<sub>2</sub> (2.7 mg/L) generated lower ethanol at any period of time. Ho et al. [32] reported that the acidic hydrolysate of *Chlorella vulgaris* produced 11.7 g/L bioethanol in separate hydrolysis and fermentation (SHF). In contrast, 79.38 g/L bioethanol was generated from the biomass of *Scenedesmus raciborskii* in fed-batch process [33]. On the whole, a similar trend is noticed with the case of ethanol production i.e. higher ethanol yield observed in cultures grown in 0.1 mg/L of nano-MnO<sub>2</sub> concentration and reduced yields were observed at elevated concentrations. Bioethanol production completely depends on the algal species, apart from different fermentation conditions. For instance, Pt-Ru/RGO nanoparticles-assisted feedstock of *C. minutum* produced 32.6 g/L of bioethanol [10]. In addition, biomass of another green alga, *Chaetomophila linum* generated 8.6 g of bioethanol per 100g dry matter, which is different from current results [34]. Overall, these findings highlight the potential of MnO<sub>2</sub> nanomaterials to modulate both algal growth and subsequent bioethanol production efficiency.

This is the first report on ethanol production by the influence of nano-MnO<sub>2</sub> in photosynthetic algal cultures. This protocol will also be useful for generating ethanol in other algal cells using nano-MnO<sub>2</sub>. Further, this work provided clues for improvement of biomass in turn ethanol production from nano-MnO<sub>2</sub>-supplemented green algae. Fig. 6 illustrates the possible role of nano-MnO<sub>2</sub> in photosynthesis, especially

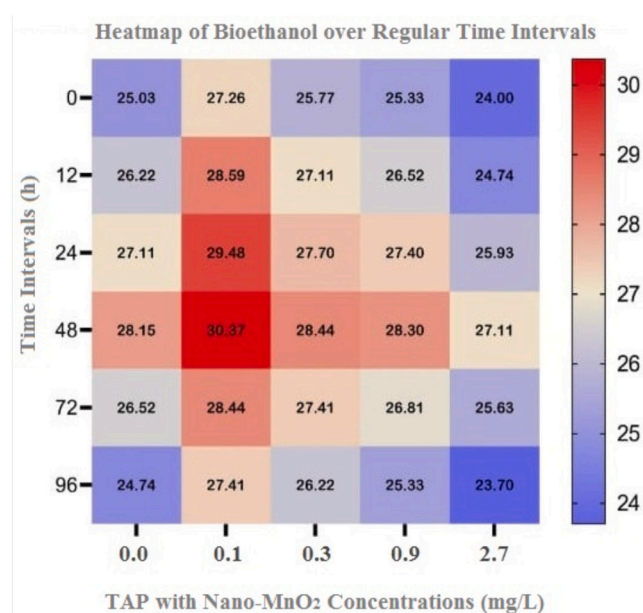
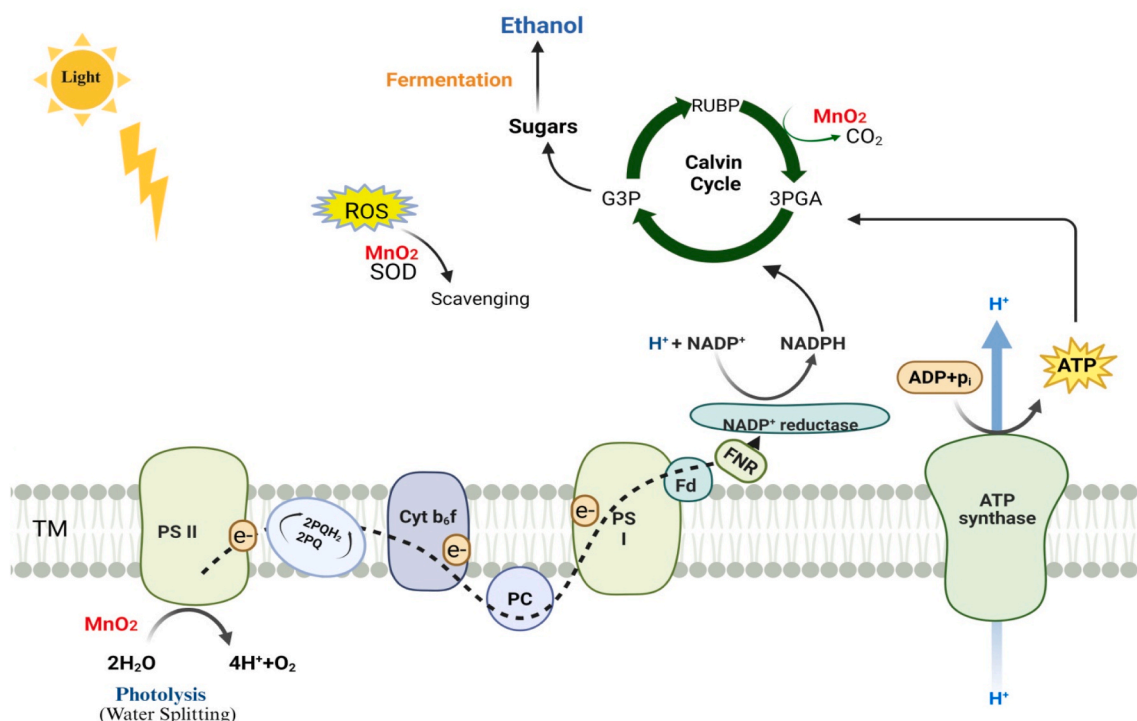


Fig. 5. Heat map of ethanol content (g/L) generated with the assistance of nano-MnO<sub>2</sub> in different time intervals.



**Fig. 6.** Possible role of nano-MnO<sub>2</sub> on photosynthesis at photosystem II and biomass enhancement in algal cells. ATP- Adenosine triphosphate, Cyt b<sub>6</sub>f- Cytochrome b<sub>6</sub>f, Fd-Ferredoxin, FNR- Ferredoxin NADP reductase, G3P- Glyceraldehyde-3-phosphate, NADPH-Nicotinamide adenine dinucleotide phosphate, PC- Plastocyanin, PQ-Plastoquinone, PSI-Photosystem I, PSII-Photosystem II, ROS-Reactive oxygen species, RUBP-Ribulose 1,5-biphosphate, SOD-Superoxide dismutase, TM-Thylakoid membrane, 3PGA-3 Phosphoglyceric acid.

in photolysis of water at photosystem II and CO<sub>2</sub> fixation, removal of stress by ROS scavenging, transport of elements and compounds in algal cells etc. Alejandro et al. [16] reported that manganese acted as a cofactor in the water-splitting phase at photosystem II. Particularly, the water-splitting mechanism needs the tetra-Mn cluster Mn<sub>4</sub>O<sub>5</sub>Ca to split two water molecules into protons, electrons and oxygen as per the works of Bricker et al. [31], suggesting the role of nano-MnO<sub>2</sub> in the improvement of biomass in turn bioethanol production. However, further research is required in depth to know the molecular reasons for the augmentation of biomass and ethanol production in algal cultures. This kind of algal work also underscores the broader environmental implications, including carbon sequestration, waste management and bioethanol production.

#### 4. Conclusions

Algae are good candidate species for various value-added products, including bioethanol. The present study emphasizes the role of nano-MnO<sub>2</sub> in improving algal biomass and ethanol production using *C. minutum*. It was noticed that TAP medium with 0.1 mg/L of nano-MnO<sub>2</sub> supplementation notably boosted algal biomass (29.40 mg/L), indicating its effectiveness in promoting growth at optimal concentration. A high amount of algal dry biomass (160 mg/L) was also observed with 0.1 mg/L of nano-MnO<sub>2</sub> treatment. The highest amount of ethanol (30.37 g/L) production was observed in feed stocks grown in TAP with 0.1 mg/L of nano-MnO<sub>2</sub> at 48 h fermentation. The presence of optimal nano-MnO<sub>2</sub> likely plays a crucial role as a catalyst in photosynthesis and CO<sub>2</sub> fixation, as well as in ROS scavenging, influencing other biochemical reactions that contribute to high levels of carbohydrate/sugar production. To the best of our knowledge, this is the first report on the role of nano-MnO<sub>2</sub> in bioethanol production from algal samples. The present investigation may be helpful to produce bioethanol at the commercial level, even using other algal cells. Overall, production of bioethanol from algal species is an environmental friendly process that dramatically

reduces our reliance on fossil fuels.

#### CRediT authorship contribution statement

**P. Gnaneswari:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **B. Sourya Swetha:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **K. Riazunnisa:** Writing – review & editing, Methodology, Formal analysis. **V. Anu Prasanna:** Writing – review & editing, Methodology, Formal analysis. **M. Rajasekhar:** Writing – review & editing, Formal analysis. **E.C. Surendranatha Reddy:** Writing – review & editing, Formal analysis. **M. Narendra Reddy:** Writing – review & editing, Formal analysis. **L. Subramanyam Sarma:** Writing – review & editing, Formal analysis. **T. Chandrasekhar:** Writing – review & editing, Writing – original draft, Supervision, Formal analysis, Conceptualization.

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#### Declaration of competing interest

The authors declared that they have no conflict of interest.

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## Data availability

The data generated during the present study is available from the corresponding author on reasonable request.

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